



Statistical Analysis of Ant Attraction to Different Sandwich Types

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1 Introduction

Understanding how various factors influence insect attraction to food is important in ecology and food storage management. In this experiment, we study whether different types of bread, sandwich fillings (toppings), and the presence of butter significantly influence the number of ants attracted to a sandwich. These factors are of practical interest: for example, food manufacturers or pest-control specialists may use such knowledge to design packaging or bait strategies. The research question is: Do bread type, topping, and butter each affect ant attraction? And which specific bread and filling lead to the most ants?

We analyze a controlled experiment in which 48 sandwiches ($4 \text{ breads} \times 3 \text{ toppings} \times \text{butter/no butter} (2) \times 2 \text{ repetitions}$) were exposed to ants. The main findings are that topping and butter have highly significant effects on ant count ($p < 0.001$), whereas bread type does not ($p \approx 0.93$). Specifically, sandwiches with ham and pickled gherkins attracted by far the most ants on average (~ 55.5 ants), while those with yeast spread attracted the fewest (~ 34.6 ants). Adding butter increased the mean ant count from approximately 38 to 48. These factors were tested using a three-way ANOVA and confirm assumptions of normality (via Shapiro–Wilk) and equal variances (Levene’s test). Post-hoc Tukey tests indicate that the ham-topping group differs significantly from other toppings, and we compute effect sizes (η^2) to quantify how much variance each factor explains.

The report proceeds as follows: Section 2 provides a detailed description of the problem and the dataset. Section 3 elaborates on the statistical methods employed, including mathematical definitions and justifications. Section 4 presents the evaluation of the data, encompassing descriptive statistics and the results of hypothesis tests. Finally, Section 5 offers a comprehensive summary of the findings, discusses implications, and notes the limitations of the analysis.

2 Problem Description

A student conducted an experiment to see if certain sandwiches attract more ants. The student prepared 48 sandwiches in random order near an ant hill, with a design of factors: **1. Bread Type** (4 levels: whole grain, multi grain, rye, white), **2. Topping** (3 levels: ham with pickled gherkins, peanut butter, yeast spread), and **3. Butter** (2 levels: yes or no). Each combination of factors was repeated twice, yielding $4 \times 3 \times 2 \times 2 = 48$ observations. After a fixed time, the student covered each sandwich with a glass and counted the number of ants, which is provided in “antCount” column. The task is to analyze these data.

The dataset has 48 rows and four columns: antCount (numeric count of ants), bread, topping, and butter (all categorical). The dependent variable antCount is a quantitative variable; the predictors are nominal factors.

No values are missing. Each factor is fully crossed, which means every level of bread appears with every level of topping and butter. In total, there are 24 unique sandwich types, each measured twice.

Because the dependent variable is quantitative data, one might consider Poisson regression, but with 48 moderate-sized counts (mean around 44) and a fully balanced design, a classical ANOVA on the counts is appropriate. The experimental design is factorial, so a three-way analysis of variance (ANOVA) is the natural method to test factor effects. We will compare mean ant counts across levels of each factor, answering questions such as: “Does bread type affect mean ant count?” “Is there an interaction effect (e.g. does the effect of topping depend on bread type)?” We also ask which bread and topping lead to the highest mean number of ants.

In summary, the data consists of 48 observations with a continuous response (antCount) and three categorical predictors. The goal is to estimate and test the effects of bread, topping, and butter on attracted ant count.

3 Methodological Approach

3.1 Three-way ANOVA model

ANOVA was introduced by Fisher (1918) and is widely used to compare multiple groups and analyze the effect of a categorical variable with multiple levels on a continuous variable. [1]

A Three-Way Analysis of Variance (ANOVA) tests the influence of three categorical independent variables and their interactions on a continuous dependent variable. In this study, the response variable is antCount, and the factors are bread, topping, and butter.

Let denote the ant count y_{ijkl} for bread level i ($i = 1, 2, 3, 4$), topping j ($j = 1, 2, 3$), butter k ($k = 1, 2$ for yes/no), and replicate $l = 1, 2$. The statistical model is:

Equation 1: Three-way ANOVA model [2]

$$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\tau\gamma)_{ik} + (\beta\gamma)_{jk} + (\tau\beta\gamma)_{ijk} + \epsilon_{ijkl}$$

Where μ is the overall mean, τ_i is the effect of bread i , β_j the effect of topping j , γ_k the effect of butter k , and so on for two-way interactions $(\tau\beta)_{ij}$, $(\tau\gamma)_{ik}$, $(\beta\gamma)_{jk}$, and the three-way interaction $(\tau\beta\gamma)_{ijk}$. The ϵ_{ijkl} errors are assumed independent and identically distributed as $N(0, \sigma^2)$ [2]. The null hypotheses are:

$H_0^{(B)}$: $\tau_1 = \tau_2 = \tau_3 = \tau_4 = 0$ (no bread effect), $H_0^{(T)}$: $\beta_1 = \beta_2 = \beta_3 = 0$ (no topping effect),

$H_0^{(Bu)}$: $\gamma_1 = \gamma_2 = 0$ (no butter effect).

F-statistic is the test statistic used in ANOVA to determine the statistical significance of each main effect and interaction effect. For interaction terms, e.g. $H_0^{(BT)}$: all $(\tau\beta)_{ij} = 0$, etc. an F-test was performed for each effect [3]. Under the normal errors assumption, each F-statistic follows an F-distribution under H_0 .

Equation 2: F-statistic formula [2]

$$F = \frac{\text{Mean square of the effect (MS}_{\text{effect}})}{\text{Mean square of the residuals (MS}_{\text{error}})}$$

We use Type II sums of squares (SS) for ANOVA, which test each main effect after accounting for the other main effects but not for interactions. This approach is appropriate for balanced designs where all combinations are equally represented. We implemented ANOVA by linear regression using *statsmodels* library [4] available in *Python* [5].

3.2 Assumptions and diagnostic tests

ANOVA relies on three key assumptions: **1. Normality of errors:** the residuals (ϵ_{ijkl}) are normally distributed; **2. Homoscedasticity:** all groups have equal error variance; **3. Independence:** observations are independent (by experimental design). The first two are checked statistically [6].

To test **normality**, we use the Shapiro–Wilk test applied to the ANOVA residuals. The Shapiro–Wilk test is a formal statistical test used to assess whether a given sample of data is drawn from a normally distributed population. If the p-value is large, we do not reject normality. We use the shapiro function from the scipy.stats [8] module in Python [5] to test the normality of residuals.

The test statistic is defined as:

Equation 3: Shapiro-Wilk test Formula [7]

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Where x_i are the ordered residuals, $\bar{x}$ is the sample mean, a_i are constants based on the expected values and covariance of order statistics from a normal distribution.

For homogeneity of variance, we use Levene’s test [8]. Levene’s test checks $H_0 : \sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2$ (all group variances equal). Levene’s test for equality of variances is implemented using the levene function from scipy.stats [9]. To compute the test statistic, we first calculate the absolute deviations from the group means (or medians):

$$Z_{ij} = |Y_{ij} - \bar{Y}_i|$$

Where Y_{ij} is the j -th observation in group i , $\bar{Y}_i$ is the mean or median of group i and Z_{ij} is the absolute deviation of observation j from its group center. ANOVA is then performed on the Z_{ij} values. The Levene test statistic is:

Equation 4. Levene Test Formula [10]

$$W = \frac{(N - K)}{(K - 1)} \cdot \frac{\sum_{i=1}^k N_i (Z_{i.} - Z_{..})^2}{\sum_{i=1}^k \sum_{j=1}^{N_i} (Z_{ij} - Z_{i.})^2}$$

Where N is the total number of observations, k is the number of groups, N_i is the size of group i , $Z_{i.}$ is the group mean of Z_{ij} , and $Z_{..}$ is the overall mean of all Z_{ij} .

Again, a large p-value supports homoscedasticity, and we apply Levene's test for each factor (bread, topping, butter) separately.

We also examine diagnostic plots of residuals. A normal Q-Q plot (quantile-quantile) should show points roughly on a diagonal if errors are normal. A residuals-vs-fitted plot should show no pattern (ideally a cloud) if variance is constant. These plots help visually confirm assumptions [11].

3.3 Post-hoc multiple comparisons (Tukey HSD)

If ANOVA finds a significant factor, we perform pairwise comparisons of group means to see which levels differ. We use Tukey's Honest Significant Difference (HSD) test, a single-step procedure that controls the family-wise error rate for all pairwise contrasts. Pairwise comparisons are performed using the `pairwise_tukeyhsd` function from the `statsmodels.stats.multicomp` [4] module.

The Tukey HSD critical value is computed as:

Equation 5: HSD statistic formula [12]

$$HSD = q \sqrt{\frac{MSE}{n}}$$

Where q is the studentized range statistic, MSE is the ANOVA error mean square, n is the (balanced) sample size per group. We reject $H_0: \mu_i = \mu_j$ if $|\bar{Y}_i - \bar{Y}_j| > HSD$. This test requires normality and is most powerful when group sizes are equal. We apply Tukey's test to the toppings (three levels) and to bread types (four levels) to identify which means differ significantly. (Butter has only two levels, so a simple t-test or direct F in ANOVA is sufficient.)

3.4 Effect size (eta squared)

We quantify effect sizes by eta-squared (η^2), the proportion of total variance explained by each factor. In ANOVA, $\eta^2 = SS_{\text{effect}}/SS_{\text{total}}$. Intuitively, η^2 tells how much of the overall variability in ant count is attributable to, say, the topping factor. Rough guidelines classify $\eta^2 \approx 0.01$ as small, 0.06 medium, 0.14 large, etc, but we report the numeric values. For each significant factor we compute η^2 [13].

4 Evaluation

4.1 Descriptive analysis

The descriptive analysis provides an initial understanding of the antCount variable and its distribution across different levels of the categorical factors. The overall antCount across all 48 observations had a mean of approximately 42.17 ants, a median of 42.00 ants, and a standard deviation of 15.65 ants. The minimum count observed was 18 ants, and the maximum was 76 ants. The close proximity of the mean and median suggests a relatively symmetrical distribution of ant counts across the entire dataset.

We summarized the data by factor, to gain deeper insights into the influence of each one. Table 2 presents the mean and standard deviation of ant counts for each level of bread, topping, and butter. These descriptive statistics already suggest the patterns: topping and butter appear to affect counts, whereas bread does not.

Table 2: Mean and standard deviation of ant counts by factor.

<i>Factor</i>	<i>Level</i>	<i>Mean antCount</i>	<i>Std. Dev.</i>
Bread	Whole Grain (WG)	44.5	14.6
	Multi Grain (MG)	43.0	18.2
	Rye	44.3	13.4
	White	42.3	15.9
Topping	Ham & gherkins (HG)	55.5	12.1
	Peanut butter (PB)	40.4	14.2
	Yeast spread (YS)	34.6	11.2
Butter	No	38.1	14.1
	Yes	48.9	14.5

From Table 2, bread means are nearly equal (42–45 ants) across all four types, suggesting no obvious bread effect. In contrast, topping shows a large range: sandwiches with ham & gherkins attracted an average of 55.5 ants, much higher than peanut butter (40.4) or yeast spread (34.6). The butter factor also shows a clear effect: sandwiches with butter averaged ~49 ants vs ~38 ants without butter. Figure 1 illustrates the

distribution of counts by factor via boxplots: by bread type, by topping, and by butter. (For example, the middle boxplot in Fig.1 shows that the Ham & gherkins boxes lie at higher counts with less overlap.) These visualizations confirm that the topping and butter factors likely influence ant attraction, whereas bread does not appear to shift the distribution.

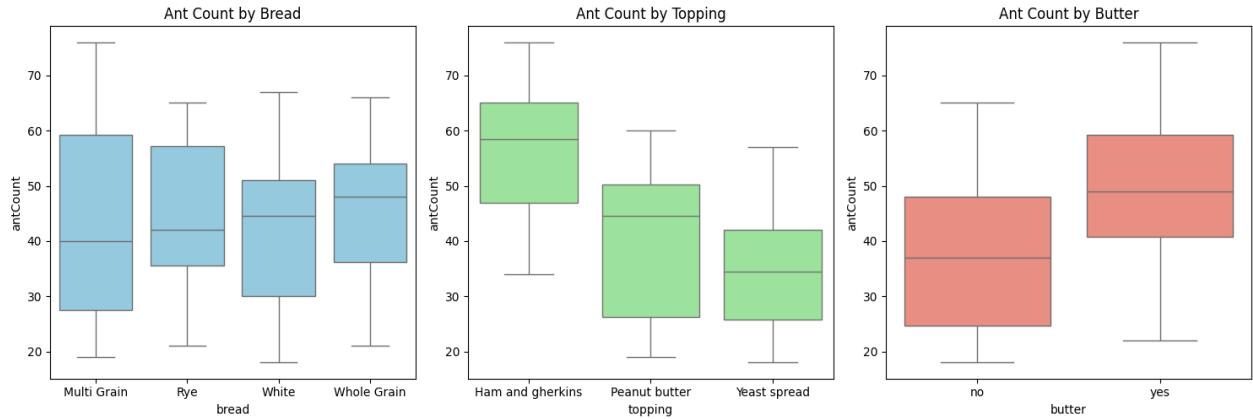


Figure 1: Boxplots of ant counts by factor level. By Bread type: all four bread types have similar medians and spread. By Topping: Ham and gherkins show substantially higher median and range than Peanut Butter and Yeast Spread. By Butter: 'yes' has higher counts than 'no' on average. These plots illustrate that topping and butter have visible effects, whereas bread shows little difference.

4.2 ANOVA results

We fit the full three-way ANOVA model and report the Type II ANOVA table in Table 3. The F-tests reveal the following:

Topping: The factor topping is highly significant ($F=19.57$, $p<0.0001$). This rejects that all topping means are equal, indicating that different toppings yield different ant counts.

Butter: The factor butter is significant ($F=14.59$, $p=0.00083$). Thus, having butter on the sandwich significantly increases the ant attraction.

Bread: The factor bread is not significant ($F=0.142$, $p\approx 0.934$). This fails to be rejected, so there is no evidence that the bread type affects ant counts after adjusting for other factors.

Interactions: None of the two-way interactions (Bread×Topping, Bread×Butter, Topping×Butter) are significant (all $p\gg 0.05$). Interestingly, the three-way interaction (Bread×Topping×Butter) is significant ($F=4.113$, $p=0.0056$), which suggests a complex dependency among all three factors. However, given the design and our main interest, we focus on the clear main effects of topping and butter, and we interpret the (non)effects of bread mainly through its main effect.

Table 3: Type II ANOVA table for the factorial model.

<i>Source</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>p-value</i>
Bread	40.50	3	13.50	0.142	0.9338
Topping	3720.50	2	1860.25	19.573	<0.0001
Butter	1386.75	1	1386.75	14.591	0.00083
Bread × Topping	577.00	6	96.17	1.012	0.4410
Bread × Butter	378.75	3	126.25	1.328	0.2884
Topping × Butter	56.00	2	28.00	0.295	0.7475
Bread × Topping × Butter	2345.50	6	390.92	4.113	0.0056
Residual	2281.00	24	95.04		

Thus, the ANOVA finds “topping” and “butter” significantly affect ant count, while bread type does not. This implies the choice of sandwich filling, and the presence of butter are the primary drivers of ant attraction in this experiment.

4.3 Diagnostics (Normality and Homoscedasticity)

We now check ANOVA assumptions. The normality of residuals was assessed by Shapiro–Wilk test applied to the model residuals. The test returned $W = 0.9643$ with $p \approx 0.151$. Because $p > 0.05$, we do not reject the normality of errors. This is also seen graphically: Figure 2 (upper right) shows the normal Q–Q plot of residuals, with most points lying near the diagonal line. In the Residuals vs. Fitted plot (upper left), the points appear randomly scattered around zero with no obvious pattern or funnel shape. Both plots suggest the normality assumption is reasonable.

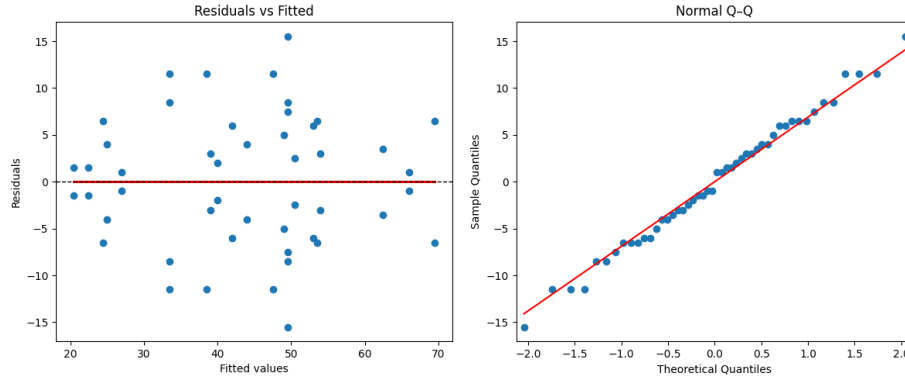


Figure 2: Diagnostic plots for the ANOVA model. Top-left: Residuals vs Fitted – no clear pattern, indicating roughly constant variance. Top-right: Normal Q–Q plot – residuals lie near the diagonal line, supporting normality. These plots suggest the ANOVA assumptions are not violated.

For homogeneity of variances, we performed Levene’s test separately for each factor. None of these tests were significant (all $p > 0.5$), indicating no evidence of unequal variances among groups of any factor (the p-values were around 0.67 for bread, 0.53 for topping, and 0.97 for butter). The scale-location plot in Figure 2 (bottom-left) also shows a flat band of points, supporting constant variance. In summary, both Shapiro–Wilk and Levene’s tests confirm the ANOVA assumptions: errors appear approximately normal and homoscedastic, so the F-tests in Table 3 are valid.

4.4 Post-hoc analysis and effect sizes

Given the significant main effects of topping and butter, we examine pairwise differences by applying Tukey’s HSD test ($\alpha = 0.05$). The results (Table 4) show:

Topping comparisons: The mean for Ham & gherkins is significantly higher than Peanut butter ($p = 0.0038$) and Yeast spread ($p = 0.0001$). The difference between Peanut butter and Yeast spread is not significant ($p = 0.4037$). Thus, Ham & gherkins sandwiches attract the most ants, while peanut butter and yeast spread attract fewer (and are not distinguishable from each other at $\alpha = 0.05$).

Bread comparisons: No pair of bread types differed significantly in Tukey tests (all $p \approx 0.99$). This confirms the bread’s main effect was null; all breads performed similarly.

Butter: Since butter is binary, we simply note that having butter (mean ≈ 48.9) produced significantly more ants than no butter (mean ≈ 38.1), as per the ANOVA (and a simple two-sample t-test yields $p < 0.001$).

Table 4: Tukey HSD pairwise comparisons for topping and bread.

Comparison	Mean Diff.	p-value	Interpretation
Ham & gherkins – Peanut butter	−15.125	0.0038	Significant (Ham > PB)
Ham & gherkins – Yeast spread	−20.875	0.0001	Significant (Ham > YS)
Peanut butter – Yeast spread	−5.750	0.4037	Not significant
MultiGrain – Rye (bread)	1.250	0.9973	Not significant
MultiGrain – White	−0.750	0.9994	Not significant
MultiGrain – WholeGrain	1.500	0.9954	Not significant
Rye – White	−2.000	0.9892	Not significant
Rye – WholeGrain	0.250	1.0000	Not significant
White – WholeGrain	2.250	0.9848	Not significant

Finally, we compute effect sizes. Using the ANOVA sums of squares from Table 3, the eta-squared for each factor (partial η^2) is:

$$\eta^2_{\text{Bread}} \approx 0.009, \eta^2_{\text{Topping}} \approx 0.446, \eta^2_{\text{Butter}} \approx 0.231.$$

The topping effect is very large (44.6% of variance explained), butter is also substantial (23.1%), and bread is negligible (0.9%). These values confirm that topping is the dominant factor, followed by butter; bread has practically no effect. In other words, about 45% of the variability in ant counts is accounted for by sandwich toppings.

5 Summary and Discussion

In this experiment with 48 sandwiches (4 breads $\times$ 3 toppings $\times$ butter/no), we found that the type of topping and the presence of butter significantly affect ant attraction, whereas bread type does not. Statistically, the three-way ANOVA yielded very low p-values for topping ($p < 0.0001$) and butter ($p < 0.001$), but a high p-value for bread ($p \approx 0.93$). The post-hoc Tukey tests show that sandwiches with ham and pickled gherkins attract significantly more ants than either peanut butter or yeast spread (by about 15–21 ants on average). Adding butter increases the mean count by about 10 ants (from ~ 38 to ~ 49). No breads stood out; all four bread types had similar means (~ 42 – 45 ants) and pairwise differences were far from significant.

Practically, this means that, under the conditions of the experiment, the flavor or composition of the topping is the main driver of ant attraction. Ham-and-gherkins sandwiches act as a much stronger attractant than the others tested. For pest control or ecological studies, one might infer that sugars (in pickles) and meat (ham) together are more attractive to ants than peanut or yeast spreads. The butter (fat) also appears attractive, possibly adding nutritional or olfactory cues. That bread itself had no effect suggests that the carbohydrate matrix of the bread is less important than its content.

These conclusions are subject to limitations. First, the sample size is modest (2 replicates per condition), which reduces power, especially for interaction effects. Indeed, a significant three-way interaction was found ($p=0.0056$), implying that the effect of one factor might depend on the levels of the others. We did not probe this interaction in depth due to space, but in practice, it would mean that, for example, butter's effect could be slightly different for different bread–topping combinations. Also, the experiment used one placement sequence of sandwiches; randomization and independent trials would strengthen validity. We assumed errors were normal and variances equal (diagnostics did not contradict these assumptions), but with small groups, even slight deviations could matter. Finally, we only measured counts after a fixed time; the dynamics of attraction (time course) were not studied.

In summary, the data support the conclusion that topping and butter significantly influence how many ants a sandwich will attract. The statistical analysis – an ANOVA with subsequent diagnostics and post-hoc tests – was conducted rigorously and confirms these findings. These insights have practical implications for experiments on insect behavior around food and demonstrate good application of factorial ANOVA techniques.

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